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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

- ☒ Probability Basics (1) (unit? unit=16&lesson=17)
- ☐ Probability Basics (2) (unit? unit=16&lesson=18)
- ☐ Linear Algebra (1) (unit? unit=16&lesson=19)
- ☐ Linear Algebra (2) (unit? unit=16&lesson=20)

Week 0: Assignment 0 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Find the eigenvalues and eigenvectors of $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix}$ **1 point**

- ☐ $2 + \sqrt{2}; (-1 - \sqrt{2}, 1, 1)^T$
- ☐ $2 - \sqrt{2}; (-1 - \sqrt{2}, 1, 1)^T$
- ☐ $2 + \sqrt{2}; (-1 + \sqrt{2}, 1, 1)^T$
- ☐ $0; (-1, 0, 1)^T$

2) Which of the following is a valid basis for the nullspace of matrix A $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix}$? **1 point**

- ☐ Empty null space
- ☐ $(-1 - \sqrt{2}, 1, 1)^T$
- ☐ $(-1 + \sqrt{2}, 1, 1)^T$
- ☐ $(-1, 0, 1)^T$

3) **1 point**

○ Optimization
(unit?
unit=16&lesso
n=21)

○ Practice:
Week 0:
Assignment 0
(Non Graded)
(assessment?
name=276)

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

Text
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Problem
Solving
Session -
July 2024 ()

The exponential of a matrix X is defined as $e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}$. Calculate e^A where A is

$$\begin{pmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{pmatrix}$$

○

$$\begin{pmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{pmatrix}$$

○

$$\begin{pmatrix} 1 & 1 & 3 \\ 8 & 6 & 15 \\ 1 & -2 & -5 \end{pmatrix}$$

○ Unbounded or infinity

○

$$\begin{pmatrix} 2 & 1 & 3 \\ 6.5 & 4.5 & 10.5 \\ -2.5 & -1.5 & -3.5 \end{pmatrix}$$

4) Which of the following matrices are orthonormal?

1 point

□

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

□

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

□

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

□

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

5) In maritime navigation, a radar system detects ships approaching a port.

1 point

Historically, 80% of the ships entering the radar's range are cargo ships, while 20% are passenger ships. It is known that cargo ships trigger the radar 95% of the time, whereas passenger ships trigger it 85% of the time. If the radar detects a ship, what is the probability that it is a cargo ship?

○ 0.35

○ 0.82

○ 0.74

○ 1

6) Say you have a random binary string generator that gives you a string of length 5.

1 point

What is the probability that the string has at least 4 zeros in it?

- ☐ 0.1875
☐ 0.2475
☐ 0.50
☐ 0.175

7) You have a bag of chocolates, with 5 of chocolate type A and 3 of chocolate type B **1 point** and 2 of chocolate type C. What is the expected value of the number of distinct chocolates you will pick up if you pick 3 chocolates (with replacement?)

- ☐ 2.02
☐ 2.12
☐ 2
☐ None of the Above

8) Let the heights and weights of 5 students in a classroom be 160, 175, 168, 155, 180 (in cm) and 60, 70, 65, 55, 75 (in kgs) respectively. Calculate the covariance between the height and weight. **1 point**

- ☐ 65
☐ 60
☐ -60
☐ 20

9) Which of the following functions are convex?

1 point

- ☐ $f(x) = x^2$
☐ $f(x) = -x^2$
☐ $f(x) = \sin(x)$
☐ $f(x) = 0$

10) Solve the following constrained optimization problem: Maximise $f(x, y) = 3x + 2y$ s. t. $x + y \leq 10$. **1 point**

- ☐ 20
☐ 30
☐ inf
☐ 0

Check Answers and Submit

